Chapter 8 — Every Lecture Example, Solved

black = write it down · red = the answer · teal = my note, do not copy · grey = optional

Chapter 8 runs from **Lec 9 slide 10** to **Lec 17 slide 19**. These are the eight places where the professor works a concrete example rather than building general theory. Each one is solved the way he solves it, with the steps he does on the board filled in.

Ex 1 TE modes in a rectangular waveguide — fields, cutoff, lowest mode — Lec 11 s6–9

Ex 2 TM modes in the same guide — why the indices change — Lec 11 s10–11

Ex 3 Attenuation in a rectangular waveguide — the C(a,b) brace and β for TE_{10} — Lec 12 s14–17

Ex 4 Cylindrical resonant cavity — full TE solution, then TM — Lec 14 s3–13

Ex 5 Resonant frequencies of that cavity — 2.405, 1.841, and the $d > 2.03R$ crossover — Lec 14 s14–17

Ex 6 Stored energy ⟨U⟩ in a cavity — Lec 15 s9–11

Ex 7 Equations for H_z and E_z in a fibre — why the modes become hybrid — Lec 17 s12–15

Ex 8 Modes in a step-index circular fibre — the eigenvalue equation and V = 2.405 — Lec 17 s16–19

All numbers checked. The attenuation bracket collapses to $a + 2b/x^2$ exactly; the minimum of Cβ sits at $\omega/\omega_\lambda = 1.8824$ for $b/a = 0.1$, which is the 1.882 on his plot. Bessel roots: $x_{01} = 2.404826$, $x'_{11} = 1.841184$, $\pi^2/x'^2_{11} = 2.9114$, crossover at $d = 2.0308\,R$. The ⟨U⟩ formula and Green's identity were checked by direct integration (10^{-16}).

Example 1 — TE modes in a rectangular waveguide

A hollow guide of perfect conductor has a rectangular cross section, sides **a** along x and **b** along y, filled with μ, ε. Find the **TE** modes, their cutoff frequencies, the lowest mode, and write out its fields. (Lec 11 s6–9.)

ANSWER

$$\psi = H_z = H_0 \cos\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right), \qquad \gamma_{mn}^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$\omega_{mn} = \frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}} \qquad m, n = 0, 1, 2\ldots \text{ but not both } 0 \quad \Rightarrow \quad \textbf{lowest TE}_{10}\textbf{:}\ \omega_{10} = \frac{\pi}{a\sqrt{\mu\varepsilon}}$$

Flow: TE means everything is built from H_z → solve the 2D wave equation for it → the wall condition for TE is $\partial\psi/\partial n = 0$, which picks **cosines** → the box quantises m and n → cutoff is where k stops being real.

Step 1 — What we start from

TE means $E_z = 0$, and the one scalar is $\psi = H_z$. With everything going as $e^{i(kz-\omega t)}$:

$$\nabla_t^2\psi + \gamma^2\psi = 0, \qquad \gamma^2 = \mu\varepsilon\omega^2 - k^2, \qquad \frac{\partial\psi}{\partial n} = 0 \text{ on the walls} \quad \text{(Lec 10, s18)}$$

$$\vec{H}_t = \frac{ik}{\gamma^2}\nabla_t\psi, \qquad \vec{E}_t = -\frac{i\omega\mu}{\gamma^2}(\hat{z}\times\nabla_t\psi) \quad \textit{needed in Step 5}$$

Step 2 — Separate x from y

Put $\psi = X(x)Y(y)$ into the wave equation and divide by XY:

$$\frac{X''}{X} + \frac{Y''}{Y} = -\gamma^2$$

The first piece depends only on x and the second only on y, yet they add to a constant. The only way is for each to be its own constant:

$$X'' = -k_x^2 X, \qquad Y'' = -k_y^2 Y, \qquad k_x^2 + k_y^2 = \gamma^2$$

Step 3 — Apply the TE wall condition

On the walls $x = 0$ and $x = a$ the outward normal is $\pm\hat{x}$, so $\partial\psi/\partial n$ there means $\partial\psi/\partial x$:

$$X'(0) = X'(a) = 0$$

X is a sine or a cosine, so X' is the other one. $X'(0) = 0$ kills the sine (its derivative is a cosine, which is 1 at 0), leaving $X = \cos(k_x x)$. Then $X'(a) = -k_x \sin(k_x a) = 0$ forces $k_x a = m\pi$. Identically for y:

$$\psi_{mn} = H_0 \cos\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right), \qquad \gamma_{mn}^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

Watch his slide here. *Lec 11 s6 lists the wall conditions as $\partial H_z/\partial x = 0$ on $y = 0, b$ and $\partial H_z/\partial y = 0$ on $x = 0, a$. Those two are* ***swapped*** *— the normal on $x = 0, a$ is $\hat{x}$, so it is $\partial H_z/\partial x$ that vanishes there. His final ψ_{mn} is right, because the cosines are symmetric in the two directions and the slip cancels.*

TE gives cosines, TM gives sines. *That one difference is the whole reason the two mode families have different allowed indices and different lowest modes. Keep it straight and Examples 1 and 2 are the same problem.*

Step 4 — Cutoff frequency, and which (m,n) are allowed

Solve $\gamma^2 = \mu\varepsilon\omega^2 - k^2$ for k:

$$k = \sqrt{\mu\varepsilon\omega^2 - \gamma_{mn}^2}$$

For the wave to travel instead of die away, k must be **real**. That needs $\mu\varepsilon\omega^2 \geq \gamma_{mn}^2$, so each mode has its own threshold frequency:

$$\omega_{mn} = \frac{\gamma_{mn}}{\sqrt{\mu\varepsilon}} = \frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}}$$ *below this the mode does not propagate*

Which pairs (m,n) are real modes? Only $m = n = 0$ has to be thrown away, because then $\psi = H_0$ is a constant, so $\nabla_t\psi = 0$ and from Step 1 every transverse field is zero — nothing is being carried.

$m, n = 0, 1, 2\ldots$ **but not both zero**

So with $a > b$ the smallest ω_{mn} is $m = 1, n = 0$:

$$\text{TE}_{10}\text{:}\quad \omega_{10} = \frac{\pi}{a\sqrt{\mu\varepsilon}} = \frac{\pi}{a}\frac{c}{n}$$ *the dominant mode; $n = c\sqrt{\mu\varepsilon}$ is the index of the filling*

Step 5 — Write out the TE_{10} fields

Put $m = 1$, $n = 0$ into Step 3: $\psi = H_0\cos(\pi x/a)$, so $\nabla_t\psi = -H_0(\pi/a)\sin(\pi x/a)\,\hat{x}$ and $\gamma^2 = (\pi/a)^2$. Feed that into the two relations of Step 1:

ANSWER — TE_{10} fields

$$H_z = H_0\cos(\pi x/a)$$

$$H_x = -i(ka/\pi)\,H_0\sin(\pi x/a)$$

$$E_y = +i(\omega\mu a/\pi)\,H_0\sin(\pi x/a)$$

$$H_y = E_x = E_z = 0$$

all $\times\, e^{i(kz-\omega t)}$, with $k = (\mu\varepsilon\omega^2 - \pi^2/a^2)^{1/2}$.

Check the walls. On $x = 0$ and $x = a$ the tangential E is E_y and E_z: E_y carries $\sin(\pi x/a)$ which is zero there, and $E_z = 0$ anyway. On $y = 0$ and $y = b$ the tangential E is E_x and E_z, both zero everywhere. And $\hat{n}\cdot\mathbf{H} = 0$: on the x walls it is H_x (carries sin, zero), on the y walls it is $H_y = 0$.

Verified numerically: these fields satisfy both Maxwell curl equations to one part in 10^8 (which is just the finite-difference error of the check itself).

■ END OF EXAMPLE 1 ■

Example 2 — TM modes in the same rectangular guide

> Same guide. Now find the **TM** modes, their cutoff frequencies and the lowest one. (Lec 11 s10–11.)

ANSWER

$$\psi = E_z = E_0 \sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right), \qquad m, n = 1, 2, 3\ldots$$

$$\omega_{mn} = \frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}} \quad \Rightarrow \quad \textbf{lowest } TM_{11}\textbf{:}\ \ \omega_{11} = \frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$$

Flow: identical to Example 1 except for **one line** — the wall condition. TM has $\psi = 0$ instead of $\partial\psi/\partial n = 0$, and everything downstream follows from that.

Step 1 — The only thing that changes

TM means $H_z = 0$ and $\psi = E_z$. The wave equation is the same, but E_z lies **in** the wall, so it is tangential and must vanish there:

$$\nabla_t^2\psi + \gamma^2\psi = 0, \qquad \psi = 0 \text{ on the walls} \quad \text{(Lec 10, s17)}$$

$$\vec{E}_t = \frac{ik}{\gamma^2}\nabla_t\psi, \qquad \vec{H}_t = \frac{i\omega\varepsilon}{\gamma^2}\,\hat{(z}\times\nabla_t\psi)$$

Step 2 — Separate, then apply $\psi = 0$

The separation is word-for-word Step 2 of Example 1: $X'' = -k_x^2 X$, $Y'' = -k_y^2 Y$. Now $X(0) = 0$ rules out the **cosine** this time, leaving $X = \sin(k_x x)$, and $X(a) = 0$ forces $k_x a = m\pi$:

$$\psi_{mn} = E_0 \sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right), \qquad \gamma_{mn}^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

Step 3 — Allowed indices and the lowest mode

$\sin(0) = 0$, so $m = 0$ or $n = 0$ wipes ψ out completely. Unlike TE, **both** indices must start at 1:

$$m, n = 1, 2, 3\ldots \quad \Rightarrow \quad TM_{11}\text{:}\ \ \omega_{11} = \frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$$

The point of the pair. γ_{mn}^2 and ω_{mn} are the same formula for TE and TM. What differs is only which (m,n) survive. TE can use $m = 0$, TM cannot, so $\omega_{TE,lowest} < \omega_{TM,lowest}$ always — which is the line he ends on. For $a = 2.5$ cm, $b = 1.2$ cm it is 2.31× higher.

■ END OF EXAMPLE 2 ■

Example 3 — Attenuation in a rectangular waveguide

> The walls are a good conductor, not a perfect one, so they absorb power. For the **TE** modes of the rectangular guide work out the two wall integrals, write −dP/dz and ⟨P⟩, and get the attenuation constant β for $\mathbf{TE_{10}}$. (Lec 12 s14–17.)

ANSWER

$$-\frac{dP}{dz} = \frac{1}{2\sigma\delta}\,\frac{\omega^2}{\omega_\lambda^2}\left\{\frac{1}{\mu\varepsilon\omega_\lambda^2}\left(1-\frac{\omega_\lambda^2}{\omega^2}\right)\left(H_o^2\frac{n^2\pi^2}{b}+H_o^2\frac{m^2\pi^2}{a}\right)+\frac{\omega_\lambda^2}{\omega^2}H_o^2 C(a,b)\right\}$$

$$TE_{10}:\ \beta = \frac{\sqrt{\varepsilon/\mu}\,(1/\sigma\delta)\,(a+2b\,\omega_\lambda^2/\omega^2)}{ab\,(1-\omega_\lambda^2/\omega^2)^{1/2}}$$

Flow: take the general TE loss formula → it needs two integrals around the rectangle, so do them wall by wall → divide by the power the guide carries → put in m = 1, n = 0.

Step 1 — What we have to compute

The general TE result (this is Jackson 8.59, derived in HW 6):

$$-\frac{dP}{dz} = \frac{1}{2\sigma\delta}\left(\frac{\omega}{\omega_\lambda}\right)^2 \oint_C \left[\frac{1}{\mu\varepsilon\omega_\lambda^2}\left(1-\frac{\omega_\lambda^2}{\omega^2}\right)\left|\hat{n}\times\nabla_t\psi\right|^2+\frac{\omega_\lambda^2}{\omega^2}\left|\psi\right|^2\right]dl$$ Lec 13, s2

So there are exactly **two** contour integrals to do around the rectangle, using $\psi_{mn} = H_o\cos(m\pi x/a)\cos(n\pi y/b)$ from Example 1. Also, for this guide

$$\omega_\lambda^2 = \frac{\pi^2}{\mu\varepsilon}\left(\frac{m^2}{a^2}+\frac{n^2}{b^2}\right) \quad = \gamma^2/\mu\varepsilon, \text{ from Example 1}$$

Step 2 — The first integrand, wall by wall

On $x = 0$ and $x = a$ the normal is $\hat{n} = \pm\hat{x}$. Crossing $\hat{x}$ into $\nabla_t\psi$ kills the x part and turns the y part into a z vector:

$$\hat{x}\times\left(\frac{\partial\psi}{\partial x}\hat{x} + \frac{\partial\psi}{\partial y}\hat{y}\right) = \frac{\partial\psi}{\partial y}\hat{z} \quad\Rightarrow\quad \left|\hat{n}\times\nabla_t\psi\right|^2 = \left|\frac{\partial\psi}{\partial y}\right|^2$$

At $x = 0$ and $x = a$ the factor $\cos(m\pi x/a)$ is $+1$ or ±1, so its square is 1 either way, and

$$\left|\hat{n}\times\nabla_t\psi\right|^2 = \left(H_0\frac{n\pi}{b}\sin\frac{n\pi y}{b}\right)^2 \qquad \text{on } x = 0,\ a$$

On $y = 0$ and $y = b$ swap the roles of x and y:

$$\left|\hat{n}\times\nabla_t\psi\right|^2 = \left(H_0\frac{m\pi}{a}\sin\frac{m\pi x}{a}\right)^2 \qquad \text{on } y = 0,\ b$$

Step 3 — Go around the rectangle

Each pair of opposite walls contributes twice the same integral, and $\int_0^b \sin^2(n\pi y/b)\,dy = b/2$:

$$\oint\left|\hat{n}\times\nabla_t\psi\right|^2 dl = 2H_0^2\frac{n^2\pi^2}{b^2}\frac{b}{2} + 2H_0^2\frac{m^2\pi^2}{a^2}\frac{a}{2} = H_0^2\left(\frac{n^2\pi^2}{b} + \frac{m^2\pi^2}{a}\right)$$

Step 4 — The second integral, and where the brace comes from

Now $\oint|\psi|^2 dl$. On $x = 0$ and $x = a$ the cosine in x is ± 1, so $|\psi|^2 = H_0^2\cos^2(n\pi y/b)$ and we need $\int_0^b \cos^2(n\pi y/b)dy$. **This is the one place where n = 0 behaves differently**, because $\cos(0) = 1$ does not average to a half:

walls	n (or m) > 0	n (or m) = 0
$x = 0$ and $x = a$	$2 \times H_0^2(b/2) = H_0^2 b$	$2 \times H_0^2 b = 2H_0^2 b$
$y = 0$ and $y = b$	$2 \times H_0^2(a/2) = H_0^2 a$	$2 \times H_0^2 a = 2H_0^2 a$

Adding the two rows for each case gives exactly his brace:

$$\oint|\psi|^2 dl = H_0^2\, C(a,b), \qquad C(a,b) = \begin{cases} a + b & m > 0,\ n > 0 \\ 2a + b & m = 0 \\ a + 2b & n = 0 \end{cases}$$

Why the doubling. With n = 0 the x walls see $|\psi|^2 = H_0^2$ flat along the whole wall instead of a $\cos^2$ that averages to one half — so that pair of walls contributes twice as much, and b becomes 2b. Same story for a when m = 0.

Putting Steps 3 and 4 into Step 1 gives the boxed $-dP/dz$ on the first page of this example.

Step 5 — The power the guide actually carries

The general TE power result:

$$\langle P\rangle = \frac{1}{2\sqrt{\mu\varepsilon}}\left(\frac{\omega}{\omega_\lambda}\right)^2\left(1-\frac{\omega_\lambda^2}{\omega^2}\right)^{1/2}\mu\int\psi^*\psi\, da$$ Lec 11, s17

The area integral of $\cos^2\cos^2$ over the rectangle is a half times a half times ab when both indices are > 0, and one of the halves becomes 1 when an index is zero:

$$\int\psi^*\psi\, da = H_0^2\begin{cases} ab/4 & m > 0,\ n > 0 \\ ab/2 & m = 0 \text{ or } n = 0 \end{cases}$$ Lec 12, s16

Step 6 — Turn loss rate into an attenuation constant

If the power falls as $P(z) = P_0 e^{-2\beta z}$, differentiate: $-dP/dz = 2\beta P$, so

$$\beta = \frac{-dP/dz}{2P}$$

Set **m = 1, n = 0** (TE_{10}). From Step 3 the first integral is $H_0^2\pi^2/a$, from Step 4 $C = a + 2b$, from Step 5 $\int\psi^*\psi\,da = H_0^2 ab/2$. Also $\omega_\lambda^2 = \pi^2/(\mu\varepsilon a^2)$, so the first bracket term becomes

$$\frac{1}{\mu\varepsilon\omega_\lambda^2}\cdot H_0^2\frac{\pi^2}{a} = \frac{a^2}{\pi^2}\cdot H_0^2\frac{\pi^2}{a} = H_0^2 a \qquad \textit{everything cancels except } a$$

So, dividing and cancelling the common $(\omega/\omega_\lambda)^2 H_0^2$:

ANSWER — β for TE_{10}

$$\beta = \frac{\sqrt{\varepsilon/\mu}\;\frac{1}{\sigma\delta}\;\left[a\left(1-\frac{\omega_\lambda^2}{\omega^2}\right)+\frac{\omega_\lambda^2}{\omega^2}(a+2b)\right]}{ab\left(1-\frac{\omega_\lambda^2}{\omega^2}\right)^{1/2}}$$

$\omega_\lambda^2/\omega^2$ may also be written $\pi^2/(a^2\mu\varepsilon\omega^2)$.

Step 7 — Tidy the bracket, and why there is a best frequency

Write $x = \omega/\omega_\lambda$. The bracket collapses:

$$a\left(1-\frac{1}{x^2}\right)+\frac{1}{x^2}(a+2b) = a - \frac{a}{x^2} + \frac{a}{x^2} + \frac{2b}{x^2} = a + \frac{2b}{x^2} \qquad \textit{the } a/x^2 \textit{ terms cancel}$$

Now the frequency dependence is visible. The skin depth itself depends on ω:

$$\delta = \sqrt{\frac{2}{\mu_c\sigma\omega}} \quad\Rightarrow\quad \frac{1}{\delta} \propto \sqrt{\omega} \propto \sqrt{x}$$

$$\Rightarrow\quad \beta \propto \frac{\sqrt{x}\,(a + 2b/x^2)}{(1 - 1/x^2)^{1/2}}$$

Read the two ends. Just above cutoff ($x \to 1$) the denominator $\to 0$, so β blows up. Far above cutoff the $\sqrt{x}$ on top wins and β grows again. ***In between there is a minimum*** *— that is the dip on his plot. Solving $d\beta/dx = 0$ numerically: the minimum sits at* ***$\omega/\omega_\lambda = 1.8824$*** *when $b/a = 0.1$, which is the 1.882 marked on the slide. For a squarer guide it moves right: 2.41 at $b/a = 0.5$, and 2.96 at $b/a = 1$.*

■ END OF EXAMPLE 3 ■

Example 4 — Cylindrical resonant cavity

> A cylinder of perfect conductor, radius **R**, length **d**, filled with μ, ε. Find the **TE** fields in full, then the TM ones. (Lec 14 s3–13.)

ANSWER — TE

$$H_z = \psi(x,y)\sin\left(\frac{p\pi z}{d}\right), \quad \vec{H}_t = \frac{p\pi/d}{\gamma^2}\cos\left(\frac{p\pi z}{d}\right)\nabla_t\psi, \quad \vec{E}_t = -\frac{i\mu\omega}{\gamma^2}\sin\left(\frac{p\pi z}{d}\right)(\hat{z}\times\nabla_t\psi)$$

with $\gamma^2 = \mu\varepsilon\omega^2 - (p\pi/d)^2$, $\psi = H_m J_m(\gamma\rho)e^{\pm im\phi}$, and everything $\times e^{-i\omega t}$.

Flow: write the two things that define the problem (Helmholtz plus the wall conditions) → translate the wall conditions into statements about H_z and E_z → solve for H_z alone, splitting z off from the cross-section → then grind the transverse fields out of the Maxwell component equations.

Step 0 — Set the problem up

Inside the cavity there are no sources, so each field obeys Helmholtz:

$$\nabla^2\vec{E} + \mu\varepsilon\omega^2\vec{E} = 0, \qquad \nabla^2\vec{H} + \mu\varepsilon\omega^2\vec{H} = 0 \qquad \text{time dependence } e^{-i\omega t}$$

The walls are perfect conductor, so on the whole surface S

$$\vec{H}_\perp = 0, \qquad \vec{E}_\parallel = 0 \quad \text{on } S, \qquad S = \begin{cases} \rho = R & \text{the side} \\ z = 0,\, d & \text{the two caps} \end{cases}$$

Step 1 — Translate the wall conditions

"Tangential E vanishes" says different things on the side and on the caps, because what counts as tangential changes. Going through it face by face:

mode	on $\rho = R$	on $z = 0, d$
TM ($H_z = 0$)	$E_z = 0$	$\mathbf{E}_t = 0$
TE ($E_z = 0$)	$\partial H_z/\partial n = 0$	$H_z = 0$ and $\mathbf{E}_t = 0$

Reading the table. On the side wall E_z runs along the wall, so it is tangential and must die — that is the TM line. For TE there is no E_z, and the condition that survives is the one on H_z: $\hat{n}\cdot\mathbf{H} = 0$ on the side forces $H_\rho = 0$, and $H_\rho \propto \partial H_z/\partial\rho$, which is why the TE side condition is a derivative. On the caps, $n = z$, so H_z is the normal component — hence $H_z = 0$ there — while E_x, E_y are the tangential ones.

Step 2 — The six component equations

These come straight from splitting the two Maxwell curl equations into a z part and a transverse part. Nothing is assumed about the z dependence — that is why they work in a cavity and not only in a guide:

① $\dfrac{\partial \vec{E}_t}{\partial z} + i\omega \hat{z}\times\vec{\mathcal{B}}_t = \nabla_t E_z$ ② $\hat{z}\cdot(\nabla_t\times\vec{E}_t) = i\omega\mathcal{B}_z$

③ $\dfrac{\partial \vec{\mathcal{B}}_t}{\partial z} - i\mu\varepsilon\omega\hat{z}\times\vec{E}_t = \nabla_t \mathcal{B}_z$ ④ $\hat{z}\cdot(\nabla_t\times\vec{\mathcal{B}}_t) = -i\mu\varepsilon\omega E_z$

⑤ $\nabla_t\cdot\vec{E}_t = -\dfrac{\partial E_z}{\partial z}$ ⑥ $\nabla_t\cdot\vec{\mathcal{B}}_t = -\dfrac{\partial \mathcal{B}_z}{\partial z}$

with $\mathcal{B} = \mu\mathbf{H}$. Lec 14, s5

Step 3 — TE: find H_z first

For TE we have $E_z = 0$, so H_z is the only z component left. It obeys Helmholtz, and from Step 1 it vanishes on both caps:

$$\nabla^2 H_z + \omega^2 \mu\varepsilon H_z = 0, \qquad H_z = 0 \text{ at } z = 0, d$$

Separate the z direction from the cross-section:

$$H_z = \psi(x,y)\, Z(z)\, e^{-i\omega t} \quad \Rightarrow \quad \nabla_t^2(\psi Z) + \frac{\partial^2}{\partial z^2}(\psi Z) + \omega^2 \mu\varepsilon(\psi Z) = 0$$

Divide through by ψZ:

$$\frac{1}{\psi}\nabla_t^2 \psi + \frac{1}{Z}\frac{\partial^2 Z}{\partial z^2} + \omega^2 \mu\varepsilon = 0, \qquad \frac{1}{Z}\frac{\partial^2 Z}{\partial z^2} \equiv -k^2$$

The middle term depends only on z while the others do not, so it must be a constant; call it $-k^2$. Then $Z'' = -k^2 Z$ with $Z(0) = Z(d) = 0$. The cosine cannot satisfy $Z(0) = 0$, so

$$Z = \sin\left(\frac{p\pi z}{d}\right), \qquad k = \frac{p\pi}{d}, \qquad p = 1, 2, 3\ldots \qquad p = 0 \text{ would kill } H_z \text{ entirely}$$

Step 4 — TE: the cross-section shape

With Z fixed, what is left of Helmholtz is a 2D equation for ψ:

$$\nabla_t^2 \psi + (\omega^2 \mu\varepsilon - k^2)\psi = 0, \qquad \gamma^2 \equiv \omega^2 \mu\varepsilon - k^2, \qquad \frac{\partial \psi}{\partial n} = 0 \text{ at } \rho = R$$

In polar coordinates the bounded solutions of this are Bessel functions:

$$\psi = H_0 J_m(\gamma_{mn}\rho)\, e^{\pm im\phi}$$ the second kind Y_m blows up at $\rho = 0$, so it is dropped

Which γ is allowed is set by the wall and is done in Example 5. Here we only need to know ψ exists and satisfies $\nabla_t^2 \psi = -\gamma^2 \psi$.

Step 5 — TE: get H_t

Take ① and ③ and put $E_z = 0$ into them:

① $\frac{\partial \vec{E}_t}{\partial z} + i\omega \hat{z}\times\vec{B}_t = 0 \quad\Rightarrow\quad \frac{\partial \vec{E}_t}{\partial z} = -i\omega\mu\,\hat{z}\times\vec{H}_t$ *call this (a)*

Now differentiate ③ with respect to z, so that $\partial \mathbf{E}_t/\partial z$ appears and (a) can be used:

$$\mu\frac{\partial^2 \vec{H}_t}{\partial z^2} - i\mu\varepsilon\omega\hat{z}\times\frac{\partial \vec{E}_t}{\partial z} = \mu\nabla_t\frac{\partial H_z}{\partial z}$$

Substitute (a). The double cross product collapses with $\hat{z}\times(\hat{z}\times\mathbf{H}_t) = -\mathbf{H}_t$, and the two i's give $i^2 = -1$, so the sign flips twice and the term comes out positive:

$$\mu\frac{\partial^2 \vec{H}_t}{\partial z^2} + \omega^2\mu^2\varepsilon\vec{H}_t = \mu k\cos(kz)\,\nabla_t\psi \quad \textit{since } \partial H_z/\partial z = k\psi\cos(kz)$$

We still need to know how $\mathbf{H}_t$ depends on z. Equation ⑥ gives it: $\nabla_t\cdot\mathbf{H}_t = -\partial H_z/\partial z \propto \cos(kz)$, so $\mathbf{H}_t \propto \cos(kz)$ and therefore $\partial^2\mathbf{H}_t/\partial z^2 = -k^2\mathbf{H}_t$. Substituting that:

$$(-\mu k^2 + \omega^2\mu^2\varepsilon)\vec{H}_t = \mu k\cos(kz)\nabla_t\psi \quad\Rightarrow\quad \vec{H}_t = \frac{\mu k}{\omega^2\mu^2\varepsilon - \mu k^2}\cos(kz)\nabla_t\psi$$

The denominator is $\mu(\omega^2\mu\varepsilon - k^2) = \mu\gamma^2$, so the μ cancels:

$$\vec{H}_t = \frac{p\pi/d}{\gamma^2}\cos\left(\frac{p\pi z}{d}\right)\nabla_t\psi$$

Step 6 — TE: get E_t

Put the result straight back into (a):

$$\frac{\partial \vec{E}_t}{\partial z} = -i\omega\mu\,\frac{p\pi/d}{\gamma^2}\cos\left(\frac{p\pi z}{d}\right)(\hat{z}\times\nabla_t\psi)$$

Integrate in z. The factor $(p\pi/d)\cos(p\pi z/d)$ is exactly the derivative of $\sin(p\pi z/d)$, so it integrates to sin:

$$\vec{E}_t = -\frac{i\mu\omega}{\gamma^2}\sin\left(\frac{p\pi z}{d}\right)(\hat{z}\times\nabla_t\psi) \quad \textit{zero at } z = 0 \textit{ and } z = d\ \checkmark$$

The integration constant is set to zero by the cap condition $\mathbf{E}_t = 0$ at $z = 0$ — and once it is zero, the sine makes $z = d$ work for free. Both caps are satisfied by one choice.

Step 7 — Collect both families

TM runs through the identical machinery with E and H swapped and $\psi = E_z$. The one difference is the cap condition: TM needs $\mathbf{E}_t = 0$ there, not $E_z = 0$, and $\mathbf{E}_t$ turns out to carry the derivative of the z function — so TM gets a cosine where TE got a sine.

ANSWER — TE modes

$$H_z = \psi(x,y)\sin(p\pi z/d)\,e^{-i\omega t}$$

$$\mathbf{H}_t = [(p\pi/d)/\gamma^2]\cos(p\pi z/d)\,\nabla_t\psi\,e^{-i\omega t}$$

$$\mathbf{E}_t = -(i\mu\omega/\gamma^2)\sin(p\pi z/d)\,(\hat{z}\times\nabla_t\psi)\,e^{-i\omega t}$$

ANSWER — TM modes

$$E_z = \psi(x,y)\cos(p\pi z/d)\,e^{-i\omega t}$$

$$\mathbf{E}_t = -[(p\pi/d)/\gamma^2]\sin(p\pi z/d)\,\nabla_t\psi\,e^{-i\omega t}$$

$$\mathbf{H}_t = (i\varepsilon\omega/\gamma^2)\cos(p\pi z/d)\,(\hat{z}\times\nabla_t\psi)\,e^{-i\omega t}$$

$\gamma^2 = \mu\varepsilon\omega^2 - (p\pi/d)^2$ in both families; TE has $p = 1, 2, 3\ldots$, TM has $p = 0, 1, 2\ldots$

The whole example in one table:

	TE	TM
built from	$\psi = H_z$	$\psi = E_z$
side wall $\rho = R$	$\partial\psi/\partial n = 0$	$\psi = 0$
z function of ψ	$\sin(p\pi z/d)$	$\cos(p\pi z/d)$
allowed p	1, 2, 3…	0, 1, 2…

Same split as the rectangular cavity in HW 6: whichever field sits along z gets the function that does **not** have to vanish at the caps, and the transverse partner gets its derivative.

■ END OF EXAMPLE 4 ■

Example 5 — Resonant frequencies of the cylindrical cavity

> Same cavity. Apply the side-wall condition to fix γ, write ω_{mnp} for both families, and find which mode is the fundamental. (Lec 14 s14–17.)

ANSWER

$$\omega_{mnp} = \frac{1}{\sqrt{\mu\varepsilon}} \left(\frac{x_{mn}^2}{R^2} + \frac{p^2\pi^2}{d^2} \right)^{1/2} \quad \text{TM, with } x_{mn} \text{ the roots of } J_m$$

$$\omega_{TM010} = \frac{2.405}{\sqrt{\mu\varepsilon}\, R}, \qquad \omega_{TE111} = \frac{1.841}{\sqrt{\mu\varepsilon}\, R} \left(1 + 2.912 \frac{R^2}{d^2}\right)^{1/2}$$

For a long cavity, $d > 2.03\,R$, TE_{111} drops below TM_{010} and becomes the fundamental.

Flow: the side wall is the only condition not yet used — use it to quantise γ → then γ and k are both fixed, so ω is too → hunt for the smallest one in each family.

Step 1 — TM: quantise γ with $E_z = 0$ at $\rho = R$

From Example 4, $\psi = E_0 J_m(\gamma\rho) e^{\pm im\phi}$. For TM the side wall demands $\psi = 0$ there:

$$J_m(\gamma R) = 0 \quad \Rightarrow \quad \gamma R = x_{mn} \quad \Rightarrow \quad \gamma_{mn} = \frac{x_{mn}}{R}$$

where x_{mn} is the nth root of $J_m(x) = 0$:

$m = 0, 1, 2\ldots$ *how many times the field wiggles going round in φ* $n = 1, 2, 3\ldots$ *which root — how many rings in ρ*

Step 2 — TM: the frequency

Example 4 defined $\gamma^2 = \mu\varepsilon\omega^2 - (p\pi/d)^2$. Solve for ω with both pieces now quantised:

$$\omega_{mnp} = \frac{1}{\sqrt{\mu\varepsilon}} \left(\frac{x_{mn}^2}{R^2} + \frac{p^2\pi^2}{d^2} \right)^{1/2}$$

TM allows $p = 0$ (its z function is a cosine), so the lowest TM mode takes the smallest root of all, $x_{01} = 2.405$, with $p = 0$:

$$\omega_{TM010} = \frac{2.405}{\sqrt{\mu\varepsilon}\, R}$$ *set by the radius alone — d does not appear*

Step 3 — TE: the same, with primes

For TE the side condition is $\partial\psi/\partial n = 0$, i.e. the derivative of the Bessel function vanishes at the wall:

$$J'_m(\gamma R) = 0 \quad \Rightarrow \quad \gamma_{mn} = \frac{x'_{mn}}{R}, \qquad x'_{mn} = \text{nth root of } J'_m(x) = 0$$

and the frequency formula is the same with x' in place of x. But **TE needs $p \geq 1$**, because its z function is a sine.

Step 4 — The lowest TE mode is not the one you expect

For TM the smallest root belongs to $m = 0$. For TE it does **not**. Comparing the first roots:

m	x_{m1} $(J_m = 0)$	x'_{m1} $(J'_m = 0)$
0	**2.405** ← smallest	3.832
1	3.832	**1.841** ← smallest
2	5.136	3.054

So the lowest TE mode is **$m = 1$, $n = 1$, $p = 1$**:

$$\omega_{TE111} = \frac{1}{\sqrt{\mu\varepsilon}} \left(\frac{1.841^2}{R^2} + \frac{\pi^2}{d^2} \right)^{1/2}$$

Pull $1.841^2/R^2$ out of the bracket. The leftover coefficient is $\pi^2/1.841^2 = 2.912$:

$$\omega_{TE111} = \frac{1.841}{\sqrt{\mu\varepsilon}\, R} \left(1 + 2.912 \frac{R^2}{d^2}\right)^{1/2}$$

Step 5 — Which one wins

TM_{010} ignores d; TE_{111} falls as d grows. So there is a crossover. Set them equal:

$$1.841^2\left(1 + 2.912 \frac{R^2}{d^2}\right) = 2.405^2 \quad \Rightarrow \quad \frac{R^2}{d^2} = \frac{5.784 - 3.389}{9.870} = 0.2427$$

$$\Rightarrow \quad \frac{d}{R} = \frac{1}{\sqrt{0.2427}} = 2.03 \quad \Rightarrow \quad \text{when } d > 2.03\,R, \ \omega_{TE111} < \omega_{TM010}$$

In words: a **squat** cavity (short and wide) resonates first in TM_{010}; a **long** one resonates first in TE_{111}. Checked numerically, the crossover is at $d = 2.0308\,R$.

■ END OF EXAMPLE 5 ■

Example 6 — Stored energy ⟨U⟩ in a cavity

> Work out the time-averaged energy stored in a cavity mode, for TM and then TE. (Lec 15 s9–11. This is the denominator of Q, so it is needed before any Q calculation.)

ANSWER

$$\langle U\rangle = \frac{d}{4}\left\{\begin{matrix}\varepsilon & \text{TM}\\ \mu & \text{TE}\end{matrix}\right\}\left[1+\left(\frac{p\pi}{d\gamma}\right)^2\right]\int_A|\psi|^2\,da$$

Flow: in a cavity the electric and magnetic energies trade places every quarter cycle, so the total is just twice the magnetic part → for TM the magnetic field is purely transverse, which makes that the easy one to integrate → the z integral gives $d/2$ → Green's identity converts $\int|\nabla_t\psi|^2$ into $\int|\psi|^2$.

Step 1 — Why ⟨U⟩ is twice the magnetic energy

Look at the TM fields from Example 4: **E** carries $\sin(p\pi z/d)$ or cos, with no i, while $\mathbf{H}_t$ carries an explicit i. An i is a quarter-cycle delay, so when E is at its peak H is zero and the other way round. The total energy therefore never changes — it just sloshes — and equals the peak of either one:

$$U=\int\mu|\vec{H}_t|^2\,da\,dz \quad\Rightarrow\quad \langle U\rangle=\int\frac{1}{2}\mu|\vec{H}_t|^2\,da\,dz$$ Lec 15, s9

For TM, $H_z = 0$, so $\mathbf{H}_t$ is the whole magnetic field — nothing is left out.

Step 2 — Square the field

From Example 4, $\mathbf{H}_t = (i\varepsilon\omega/\gamma^2)\cos(p\pi z/d)(\hat{z}\times\nabla_t\psi)$. Crossing with a unit vector does not change length, so $|\hat{z}\times\nabla_t\psi| = |\nabla_t\psi|$:

$$|\vec{H}_t|^2=\frac{\varepsilon^2\omega^2}{\gamma^4}\cos^2\left(\frac{p\pi z}{d}\right)|\nabla_t\psi|^2$$

Step 3 — Do the z integral

Only the cosine depends on z, and $\int_0^d \cos^2(p\pi z/d)dz = d/2$ for $p > 0$:

$$\langle U\rangle = \frac{1}{2}\,\frac{\mu\varepsilon^2\omega^2}{\gamma^4}\,\frac{d}{2}\int_A |\nabla_t\psi|^2 da = \frac{d}{4}\,\frac{\mu\varepsilon^2\omega^2}{\gamma^4}\int_A |\nabla_t\psi|^2 da$$

Step 4 — Trade $\int|\nabla_t\psi|^2$ for $\int|\psi|^2$

Start from the product rule $\nabla_t\cdot(\psi^*\nabla_t\psi) = |\nabla_t\psi|^2 + \psi^*\nabla_t^2\psi$, and integrate it over the cross-section. The divergence theorem turns the left side into a boundary loop:

$$\int_A |\nabla_t\psi|^2 da = \oint_C \psi^*\frac{\partial\psi}{\partial n}\,dl - \int_A \psi^*\nabla_t^2\psi\, da \qquad \textit{Green's first identity}$$

The loop term is **zero either way**: for TM $\psi = 0$ on the wall, and for TE $\partial\psi/\partial n = 0$. And $\nabla_t^2\psi = -\gamma^2\psi$ from the wave equation, so the remaining integral flips sign:

$$\int_A |\nabla_t\psi|^2 da = \gamma^2\int_A |\psi|^2 da$$

$$\Rightarrow \langle U\rangle = \frac{d}{4}\,\frac{\mu\varepsilon^2\omega^2}{\gamma^2}\int_A |\psi|^2 da \qquad \textit{one } \gamma^2 \textit{ cancelled}$$

Step 5 — Replace $\mu\varepsilon\omega^2$

From Example 4, $\mu\varepsilon\omega^2 = \gamma^2 + k^2$ with $k = p\pi/d$. Substituting kills both ω and μ:

$$\langle U\rangle = \frac{d}{4}\,\varepsilon\,\frac{\gamma^2 + k^2}{\gamma^2}\int_A |\psi|^2 da = \frac{d}{4}\,\varepsilon\left[1 + \left(\frac{p\pi}{d\gamma}\right)^2\right]\int_A |\psi|^2 da$$

ANSWER — Example 6

TM: $\langle U\rangle = \frac{d}{4}\,\varepsilon\left[1 + \left(\frac{p\pi}{d\gamma}\right)^2\right]\int_A |\psi|^2 da$ **TE:** $\langle U\rangle = \frac{d}{4}\,\mu\left[1 + \left(\frac{p\pi}{d\gamma}\right)^2\right]\int_A |\psi|^2 da$

TE is the mirror image: there E_t is the one with the i, so the roles of ε and μ swap. The bracket is identical.

Checked by brute force on a rectangular TM_{112} cavity: both the Green's identity step and the final formula agree with direct numerical integration to 10^{-16}.

■ END OF EXAMPLE 6 ■

Example 7 — Equations for H_z and E_z in a fibre

A dielectric guide where the index **varies across the cross-section**, n = n(x,y), with no metal walls. Derive the equations E_z and H_z obey, and say what changes compared with a hollow metal guide. (Lec 17 s12–15.)

ANSWER

$$\nabla_t^2 E_z + \gamma^2 E_z - \left(\frac{k}{\gamma n}\right)^2 (\nabla_t n^2)\cdot\nabla_t E_z = \frac{\omega k \mu_0}{\gamma^2 n^2}\,\hat{z}\cdot[\nabla_t n^2 \times \nabla_t H_z]$$

$$\nabla_t^2 H_z + \gamma^2 H_z - \left(\frac{\omega}{\gamma c}\right)^2 (\nabla_t n^2)\cdot\nabla_t H_z = -\frac{\omega k \varepsilon_0}{\gamma^2}\,\hat{z}\cdot[\nabla_t n^2 \times \nabla_t E_z]$$

*$\gamma^2 = n^2\omega^2/c^2 - k^2$. These are **not** Laplace equations any more, and the two of them are **coupled** — so the modes are **hybrid**, with $E_z \neq 0$ and $H_z \neq 0$ together.*

Flow: the one new fact is that $\nabla\cdot\mathbf{E}$ is no longer zero when n varies → that leaves an extra term on the right of the wave equation → take its z component, feed in the usual transverse-field relation, and the extra term turns into the coupling.

Step 1 — The one thing that is different

There is no free charge, so $\nabla\cdot\mathbf{D} = 0$, i.e. $\nabla\cdot(\varepsilon_0 n^2 \mathbf{E}) = 0$. Expand with the product rule:

$$n^2 \nabla\cdot\vec{E} + (\nabla n^2)\cdot\vec{E} = 0 \quad\Rightarrow\quad \nabla\cdot\vec{E} = -\frac{1}{n^2}(\nabla n^2)\cdot\vec{E}$$

In a hollow metal guide n is constant, $\nabla n^2 = 0$, and this is just $\nabla\cdot\mathbf{E} = 0$. Everything that follows is the price of letting n vary.

Step 2 — The wave equation picks up a right-hand side

Take the curl of $\nabla\times\mathbf{E} = i\omega\mu_0\mathbf{H}$ and use the vector identity $\nabla\times(\nabla\times\mathbf{E}) = \nabla(\nabla\cdot\mathbf{E}) - \nabla^2\mathbf{E}$, together with $\nabla\times\mathbf{H} = -i\omega\varepsilon_0 n^2\mathbf{E}$:

$$\nabla^2\vec{E} + \frac{n^2\omega^2}{c^2}\vec{E} = \nabla(\nabla\cdot\vec{E}) = -\nabla\left(\frac{1}{n^2}(\nabla n^2)\cdot\vec{E}\right)$$

Take the z component and write everything as $e^{ikz - i\omega t}$:

$$\nabla^2 E_z + \frac{n^2\omega^2}{c^2}E_z = -\left[\nabla\left(\frac{1}{n^2}(\nabla n^2)\cdot\vec{E}\right)\right]\cdot\hat{z}$$

Step 3 — Tidy the left side

Split $\nabla^2 = \nabla_t^2 + \partial^2/\partial z^2$, and the z derivative brings down $(ik)^2 = -k^2$:

$$\text{LHS} = \nabla_t^2 E_z - k^2 E_z + \frac{n^2\omega^2}{c^2} E_z = \nabla_t^2 E_z + \gamma^2 E_z, \qquad \gamma^2 \equiv \frac{n^2\omega^2}{c^2} - k^2$$

Step 4 — Tidy the right side

Assume the index does not change along the fibre, $\partial n^2/\partial z = 0$, so $\nabla n^2 = \nabla_t n^2$. Then the outer ∇ dotted with $\hat{z}$ is just $\partial/\partial z$, and it acts only on $\mathbf{E}$:

$$\text{RHS} = -\frac{\partial}{\partial z}\left(\frac{1}{n^2}\nabla_t n^2 \cdot \vec{E}\right) = -\frac{1}{n^2}\nabla_t n^2 \cdot \frac{\partial \vec{E}}{\partial z} = -\frac{ik}{n^2}\nabla_t n^2 \cdot \vec{E}_t$$

(only $\mathbf{E}_t$ survives, because $\nabla_t n^2$ has no z component to dot with E_z).

Step 5 — Put in E_t, and the coupling appears

The usual waveguide relation still holds here — it came from Maxwell, not from the walls:

$$\vec{E}_t = \frac{i}{\gamma^2}\,[k\nabla_t E_z - \omega\mu_0 \hat{z}\times\nabla_t H_z]$$ Lec 10, s5–7

Substitute into Step 4. The two i's multiply to −1, which cancels the minus sign in front:

$$RHS = \frac{k}{\gamma^2 n^2}\,\nabla_t n^2\cdot[k\nabla_t E_z - \omega\mu_0 \hat{z}\times\nabla_t H_z]$$

Split the dot product into its two pieces:

$$= \frac{k^2}{\gamma^2 n^2}\,\nabla_t n^2\cdot\nabla_t E_z - \frac{\omega\mu_0 k}{\gamma^2 n^2}\,\nabla_t n^2\cdot(\hat{z}\times\nabla_t H_z)$$

Use the rule $\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B})$ to rotate the second triple product so that $\hat{z}$ sits in front:

$$\nabla_t n^2\cdot(\hat{z}\times\nabla_t H_z) = -\hat{z}\cdot(\nabla_t n^2\times\nabla_t H_z)$$

$$\Rightarrow\ RHS = \frac{k^2}{\gamma^2 n^2}\,\nabla_t n^2\cdot\nabla_t E_z + \frac{\omega\mu_0 k}{\gamma^2 n^2}\,\hat{z}\cdot(\nabla_t n^2\times\nabla_t H_z)$$

Move the first piece to the left (it only involves E_z) and note $k^2/(\gamma^2 n^2) = (k/\gamma n)^2$:

ANSWER — Example 7

$$\nabla_t^2 E_z + \gamma^2 E_z - \left(\frac{k}{\gamma n}\right)^2 (\nabla_t n^2)\cdot\nabla_t E_z = \frac{\omega k\mu_0}{\gamma^2 n^2}\,\hat{z}\cdot[\nabla_t n^2 \times \nabla_t H_z]$$

$$\nabla_t^2 H_z + \gamma^2 H_z - \left(\frac{\omega}{\gamma c}\right)^2 (\nabla_t n^2)\cdot\nabla_t H_z = -\frac{\omega k\varepsilon_0}{\gamma^2}\,\hat{z}\cdot[\nabla_t n^2 \times \nabla_t E_z]$$

The H_z equation comes out of the identical argument started from $\nabla\times\mathbf{H}$ instead of $\nabla\times\mathbf{E}$.

Step 6 — What this costs us

	hollow metal guide	fibre, n = n(x,y)
equation type	plain 2D Helmholtz	extra gradient term
E_z and H_z	independent	coupled
modes	pure TE or pure TM	**hybrid**: both non-zero

The coupling switches off exactly when the right-hand cross product vanishes. For a fibre with $n = n(\rho)$, $\nabla_t n^2$ points along $\hat{\rho}$; if the mode also has no ϕ dependence, $\nabla_t H_z$ points along $\hat{\rho}$ too, the cross product of two parallel vectors is zero, and the equations decouple again.

pure TE / TM only when $\psi = \psi(\rho)$, i.e. azimuthal index $m = 0$

■ END OF EXAMPLE 7 ■

Example 8 — Modes in a step-index circular fibre

A fibre with a core of index n_1 for $\rho < a$ and cladding n_2 for $\rho > a$. Find the mode shapes, the condition that picks the allowed modes, and the lowest cutoff frequency. (Lec 17 s16–19.)

ANSWER

$$\left(\frac{n_1^2}{\gamma}\frac{J'_m}{J_m} + \frac{n_2^2}{\beta}\frac{K'_m}{K_m}\right)\left(\frac{1}{\gamma}\frac{J'_m}{J_m} + \frac{1}{\beta}\frac{K'_m}{K_m}\right) = \frac{m^2}{a^2}\left(\frac{n_1^2}{\gamma^2} + \frac{n_2^2}{\beta^2}\right)\left(\frac{1}{\gamma^2} + \frac{1}{\beta^2}\right)$$

lowest TE/TM cutoff: $V = \dfrac{n_1 \omega a\sqrt{2\Delta}}{c} = 2.405 \quad \Rightarrow \quad \omega = \dfrac{2.405\,c}{n_1 a\sqrt{2\Delta}}$

Flow: the index is constant inside and constant outside, so the coupling term of Example 7 lives **only at the interface** → solve a plain Helmholtz equation on each side → the coupling comes back as the matching condition at $\rho = a$, and that is the boxed equation.

Step 1 — Where the coupling lives

Write the step profile and differentiate it. A step drops to a delta function:

$$n = \begin{cases} n_1 & \rho < a \\ n_2 & \rho > a \end{cases} \quad \Rightarrow \quad \nabla_t n^2 = -2n_1^2 \Delta\, \delta(\rho - a)\, \hat{\rho}$$

where Δ is the usual fractional index step:

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2} \quad \Rightarrow \quad n_1^2 - n_2^2 = 2n_1^2 \Delta \qquad \textit{so the jump in } n^2 \textit{ is exactly } 2n_1^2\Delta$$

This is the whole trick of a step-index fibre. Away from $\rho = a$ the gradient is zero, so all the nasty terms of Example 7 vanish and we are back to ordinary Helmholtz equations. The coupling has not disappeared — it has been squeezed into a single surface, where it becomes a matching condition.

Step 2 — Solve on each side

Inside ($\rho < a$). Guided light travels mostly in the core, so $\gamma^2 = n_1^2\omega^2/c^2 - k^2 > 0$ and the equation is oscillatory. Bounded at the axis ⇒ ordinary Bessel:

$$\left\{ \begin{matrix} E_z \\ H_z \end{matrix} \right\} = \left\{ \begin{matrix} A_e \\ A_h \end{matrix} \right\} J_m(\gamma\rho)\, e^{im\phi}$$

Outside ($\rho > a$). For the light to stay trapped the field must **decay**, so there the sign flips:

$$\beta^2 = k^2 - \frac{n_2^2\omega^2}{c^2} > 0 \quad \Rightarrow \quad \nabla_t^2\psi - \beta^2\psi = 0 \quad \Rightarrow \quad \left\{ \begin{matrix} E_z \\ H_z \end{matrix} \right\} = \left\{ \begin{matrix} B_e \\ B_h \end{matrix} \right\} K_m(\beta\rho)\, e^{im\phi}$$

*K_m is the modified Bessel function that **dies** as $\rho \to \infty$. Its partner I_m grows without bound, so it is thrown away — the same reasoning that threw away Y_m at the axis, just at the other end.*

Step 3 — Match at ρ = a

Four unknowns (A_e, A_h, B_e, B_h) and four conditions — the tangential fields E_z, H_z, E_ϕ, H_ϕ must be continuous across the interface. That is a 4×4 homogeneous system; a non-trivial solution exists only if its determinant vanishes, and that determinant is the boxed equation:

$$\left(\frac{n_1^2}{\gamma}\frac{J'_m}{J_m}+\frac{n_2^2}{\beta}\frac{K'_m}{K_m}\right)\left(\frac{1}{\gamma}\frac{J'_m}{J_m}+\frac{1}{\beta}\frac{K'_m}{K_m}\right)=\frac{m^2}{a^2}\left(\frac{n_1^2}{\gamma^2}+\frac{n_2^2}{\beta^2}\right)\left(\frac{1}{\gamma^2}+\frac{1}{\beta^2}\right)$$

Lec 17, s17

All the Bessel functions are evaluated at ρ = a.

Step 4 — Read the equation: m = 0 against m > 0

m = 0. The right side is zero, so the left side factorises and each bracket can vanish on its own:

$(n_1^2/\gamma)(J'_0/J_0) + (n_2^2/\beta)(K'_0/K_0) = 0$ **TM modes**

$(1/\gamma)(J'_0/J_0) + (1/\beta)(K'_0/K_0) = 0$ **TE modes**

m > 0. The right side is non-zero, the two brackets cannot be separated, and E_z and H_z must both be present — these are the **hybrid** (EH and HE) modes, exactly the coupling predicted in Example 7.

Rays, in picture language. *m = 0 modes are* ***meridional rays*** *— they zig-zag through the axis, staying in one plane. m > 0 modes are* ***skew rays*** *— they spiral around the axis without ever crossing it, which is why their intensity is zero at ρ = 0.*

Step 5 — Cutoff, and the number 2.405

Cutoff is the moment the mode stops being guided: the field outside stops decaying, so $\beta \to 0$. Add the two definitions of γ^2 and β^2 from Step 2 — k^2 cancels:

$$\gamma^2 + \beta^2 = \frac{(n_1^2 - n_2^2)\omega^2}{c^2} = \frac{2n_1^2\Delta\,\omega^2}{c^2}$$

At cutoff $\beta = 0$, so γ takes the whole of it. Define $\mathbf{V \equiv \gamma a}$ at that point:

$$V = \gamma a = \frac{n_1 \omega a \sqrt{2\Delta}}{c}$$ *the fibre V-number*

With $\beta = 0$ the K-terms drop out of the $m = 0$ conditions and both reduce to $J_0(V) = 0$, whose first root is 2.405:

$$V = 2.405 \quad \Rightarrow \quad \omega = \frac{2.405\,c}{n_1 a \sqrt{2\Delta}}$$

Two things worth keeping. *(i) The hybrid HE_{11} mode has* ***no cutoff*** *— it propagates at every frequency — so a fibre with $V < 2.405$ carries exactly one mode. That is the definition of a single-mode fibre. (ii) $n_1\sqrt{(2\Delta)}$ is identically $\sqrt{(n_1^2 - n_2^2)}$, the* ***numerical aperture****, so $V = NA \cdot \omega a/c$.*

■ END OF EXAMPLE 8 ■